ASSIGNMENT 7: GAN

1.

A. Explain the fundamental architecture of a GAN, describing the roles of the generator and discriminator networks. How do these two components interact during training, and what is the theoretical goal of this adversarial process?

Ans: A GAN, or Generative Adversarial Network, has two neural networks working against each other: the generator and the discriminator. The generator creates fake samples, usually by starting with random noise and turning it into something like an image. The discriminator acts like a judge, checking both real samples from the dataset and fake samples from the generator. During training, the generator tries to create outputs realistic enough to fool the discriminator, while the discriminator tries to correctly identify which samples are fake. This repeated competition helps both networks improve. The goal is for the generator to learn the real data distribution so well that the discriminator can no longer confidently differentiate between real and generated samples. Ideally, the discriminator's accuracy becomes close to 50%, since the fake samples look just like the real ones.

B. Define what mode collapse is and explain why it occurs. Discuss at least 2 techniques that have been proposed to mitigate this problem, explaining the mechanism by which each approach helps stabilize training.

Ans: Mode collapse happens when the generator keeps producing very similar outputs instead of learning the full variety present in the training data. For example, a GAN trained on thousands of human faces might repeatedly create faces with almost the same appearance. This usually happens because the generator discovers one output pattern that successfully fools the discriminator, so it keeps using that same strategy. One technique used to reduce mode collapse is mini-batch discrimination. Here, the discriminator compares multiple generated samples at the same time and checks whether they are too similar. If the samples lack variety, the generator receives feedback encouraging it to create more diverse outputs. Another approach is Wasserstein GAN, or WGAN. Instead of giving only a simple real-or-fake answer, the critic gives a continuous score showing how close the generated distribution is to the real distribution. This provides smoother gradients and more useful feedback, making training more stable and reducing the chance of the generator getting stuck on one repeated pattern. Unrolled GANs also help by allowing the generator to consider how the discriminator may change during future training steps, instead of reacting only to its current behavior.

C. Describe three significant variants or extensions of the original GAN architecture (DCGAN, StyleGAN, Conditional GAN, CycleGAN, or others of your choice). For each variant, explain what specific problem or limitation it addresses and what innovations or modifications it introduces to achieve its goals.

Ans: Several important GAN variants have come up to solve limitations of the original architecture. DCGAN, or Deep Convolutional GAN, uses convolutional layers and transposed convolutional layers, making it much better suited for image generation. It helps the model learn visual features such as edges, shapes, and textures while improving training stability. A Conditional GAN adds extra information, such as a class label or text description, to both the generator and discriminator. This gives the user more control over the output, such as asking the model to generate a specific type of object. CycleGAN focuses on translating images between two different domains, such as turning horse images into zebra images or summer scenes into winter scenes. Its main advantage: no paired images required. It uses two generators, two discriminators, and cycle-consistency loss, which checks whether an image can be translated to another domain and then converted back while still preserving its main content.